DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 5

Mobile app developer

Profession 5 of 10 in Development and IT.

What the profession means

A mobile app developer builds applications for phones and tablets. Most work uses one code base for both Android and iPhone through a cross-platform tool such as React Native (with Expo) or Flutter, or native tools (Kotlin for Android, Swift for iPhone).

What professionals do every day

Designs screens for touch, connects to data and APIs, handles notifications and permissions, tests on real devices, and prepares store submissions.

Why businesses hire them, and what problems they solve

Customers spend more time on phones than on desktops. Businesses want an app for bookings, loyalty, ordering or internal work, and a developer can deliver one code base for both stores.

- Apps that feel slow, crash or drain the battery.
- Login, payments and notifications that must work on both platforms.
- Store rejections caused by missing policies and poor testing.

Skills you need

- JavaScript or Dart (depending on the tool), and component thinking.
- Navigation, forms, lists and state.
- APIs, storage and authentication.
- Device testing, permissions and performance.
- App store rules and privacy policies.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Expo	Build React Native apps with a simple workflow	Free	Paid build services	expo.dev
React Native	Cross-platform UI framework	Free	Free	reactnative.dev
Flutter	Cross-platform UI toolkit with Dart	Free	Free	flutter.dev
Android Studio	Official Android IDE	Free	Free	developer.android.com/studio
Xcode	Official Apple development tool (needs a Mac)	Free	Free (Apple developer account for publishing has a fee)	developer.apple.com/xcode

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Screens, lists, forms and local state with a single-screen app.
Intermediate	Navigation, data from an API, storage, and testing on a real phone.
Advanced	Authentication, push notifications, offline support, performance tuning and store submission.

Projects by level

- **Beginner:** A checklist or habit tracker.
- **Intermediate:** A booking app that reads from an API.
- **Advanced:** A loyalty app with login, notifications and an admin view.

Client-ready portfolio project

A two-screen app for a real or clearly fictional local business (ordering, bookings or loyalty), tested on Android and iPhone, with a privacy policy and store-ready screenshots.

Worked example

Worked example. A list screen is a `FlatList` fed by an array. Adding an item means creating a new array, which the list redraws. Changing an item means mapping over the array and replacing one object.

Assignment with full solution

ASSIGNMENT

A habit checklist app on a phone

Business problem: A student wants a phone app to add daily study habits and tick them off. It must reject empty names and work on both Android and iPhone.

Steps

1. Create an Expo project from the official template and open it in Expo Go on your phone.
2. Replace `App.js` with the solution.
3. Add three habits, tick one, and check that empty input is ignored.
4. Close and reopen the app; note that the data is lost (this is expected in this version).

Expected output: A working checklist on your phone: add, tick and ignore empty input. Notes on what is not yet saved.

SOLUTION (ANSWER KEY)

The solution keeps habits in memory with `useState`, so an empty name is ignored by the `trim()` check. Tapping a row flips its `done` flag using an immutable update (a new array), which React needs to redraw the list. **Not run here:** this needs Expo and a phone; the code follows the Expo template conventions, but test it on a device before you rely on it. The next step, saving data, uses a storage library that the Expo documentation covers.

JAVASCRIPT

```
// App.js (React Native with Expo). Run with: npx expo start, then scan the QR code in Expo
Go.
import { useState } from 'react';
import { Button, FlatList, StyleSheet, Text, TextInput, View } from 'react-native';

export default function App() {
  const [habits, setHabits] = useState([]);
  const [name, setName] = useState('');

  const addHabit = () => {
    const clean = name.trim();
    if (!clean) return;                       // ignore empty input
    setHabits([...habits, { id: String(Date.now()), name: clean, done: false }]);
    setName('');
  };

  const toggle = (id) =>
    setHabits(habits.map((h) => (h.id === id ? { ...h, done: !h.done } : h)));

  return (
    <View style={styles.screen}>
      <Text style={styles.title}>Study habits</Text>
      <TextInput style={styles.input} value={name} onChangeText={setName}
                 placeholder="New habit, e.g. 20 minutes of SQL" />
      <Button title="Add" onPress={addHabit} />
      <FlatList data={habits} keyExtractor={(h) => h.id}
        renderItem={({ item }) => (
          <Text style={styles.row} onPress={() => toggle(item.id)}>
            {item.done ? '✓ ' : '○ '}{item.name}
          </Text>)} />
    </View>
  );
}

const styles = StyleSheet.create({
  screen: { flex: 1, padding: 24, paddingTop: 60 },
  title: { fontSize: 24, fontWeight: '700', marginBottom: 12 },
  input: { borderWidth: 1, borderColor: '#ccc', borderRadius: 8, padding: 10, marginBottom: 8
},
  row: { fontSize: 18, paddingVertical: 10 },
});
```

Common mistakes

- Testing only in an emulator; always test on a real device.
- Ignoring the store's privacy and permission rules.
- Storing secrets (API keys) inside the app bundle.
- Building without a plan for offline or slow networks.

Freelancing opportunities

- App builds for small businesses (booking, loyalty, ordering).
- Store submission and maintenance contracts.
- Search 'React Native', 'Expo', 'Flutter', 'mobile app' on freelance marketplaces; present a working demo on a phone.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- React Native: getting started
- Expo documentation
- Flutter documentation
- Android Developers courses

Part of **Master Freelancing: Development and IT**. **Back to the chapter list on skillnavora.in** · **Open the full textbook**. Official links in this chapter were not checked live; confirm before relying on them.